Thermal Physics 2

Chapter 2: Heat & Energy Transfer

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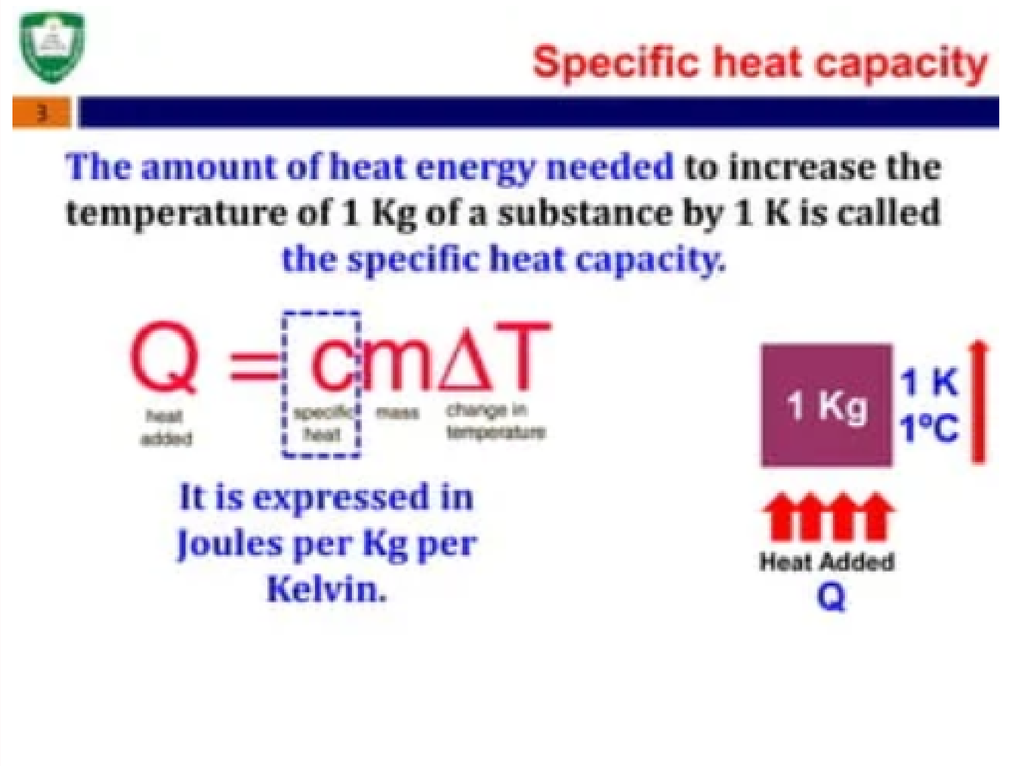
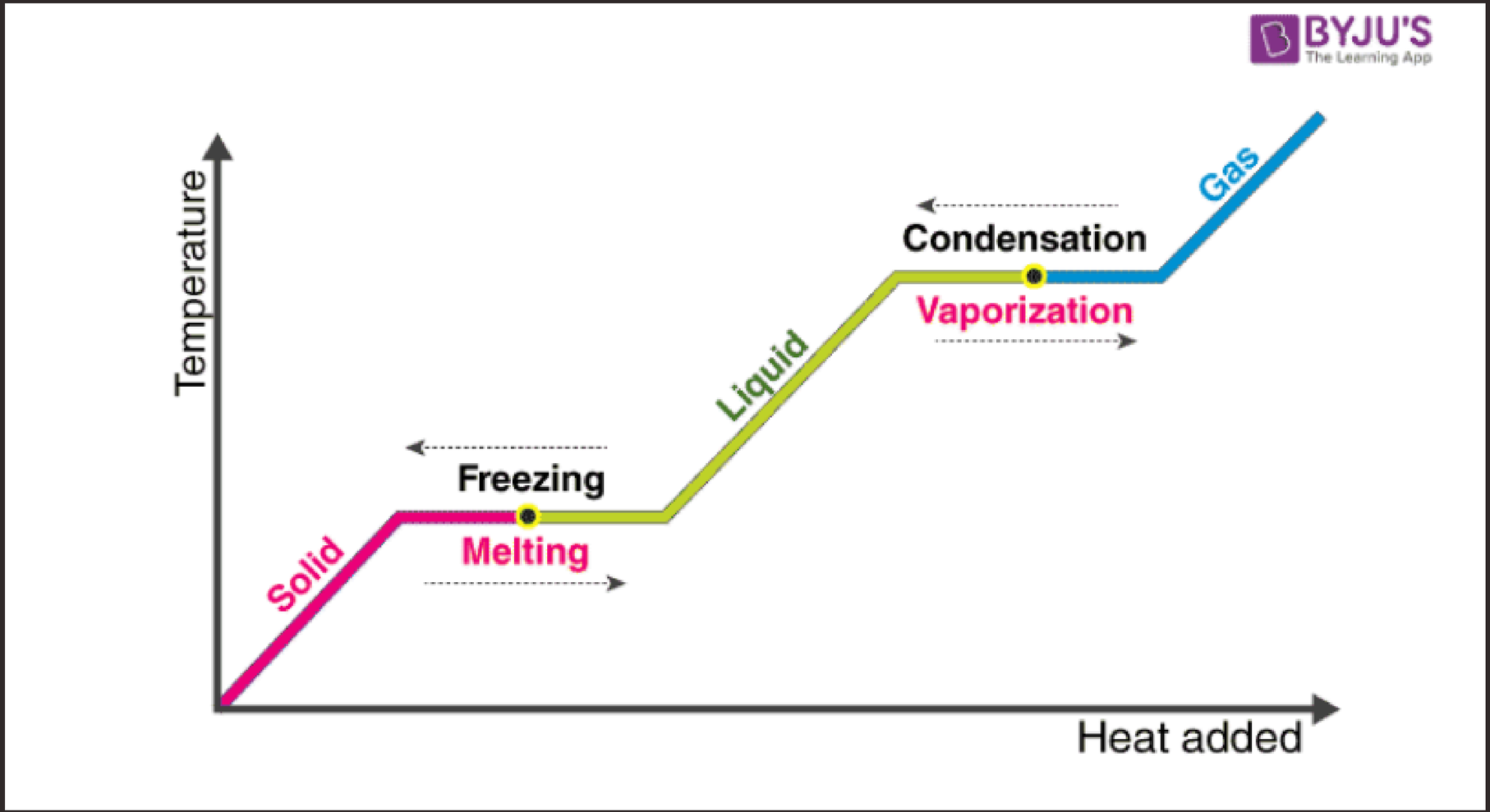
Heat Capacity & Specific Heat Capacity



Key formula:
$Q = mc\triangle T$

Where:
Q = heat energy (J)
m = mass (kg)
c = specific heat capacity
$\triangle T$ = temperature change

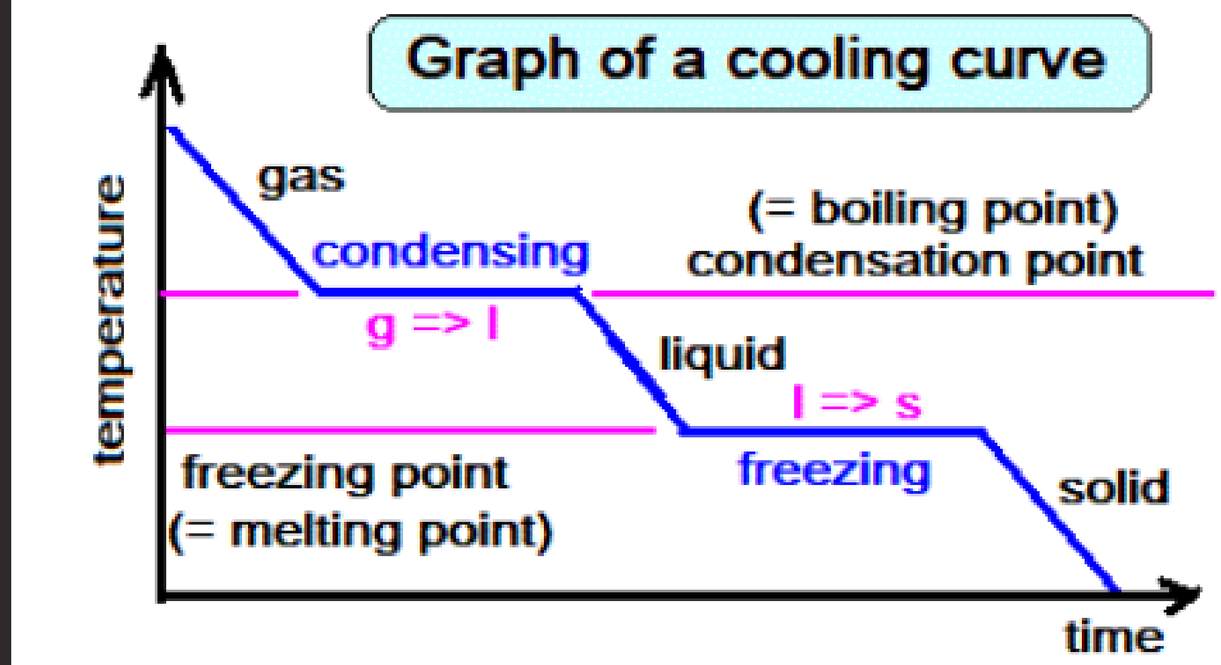
Latent Heat & Phase Changes

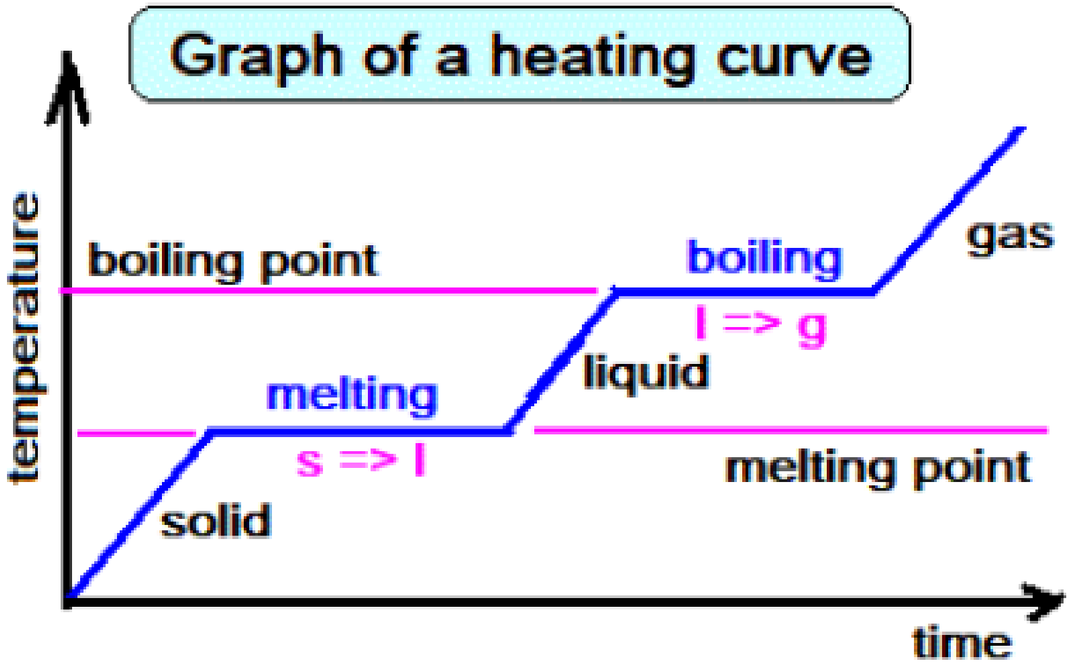


Boiling vs Evaporation

Property	Boiling	Evaporation
Temperature	Fixed (boiling point)	Any Temperature
Location	Throughout liquid	Only at surface
Speed	FAST	slow
Bubbles	Produce bubbles	No bubbles

Heating & Cooling Curves





Black’s Principle Thermal Equilibrium

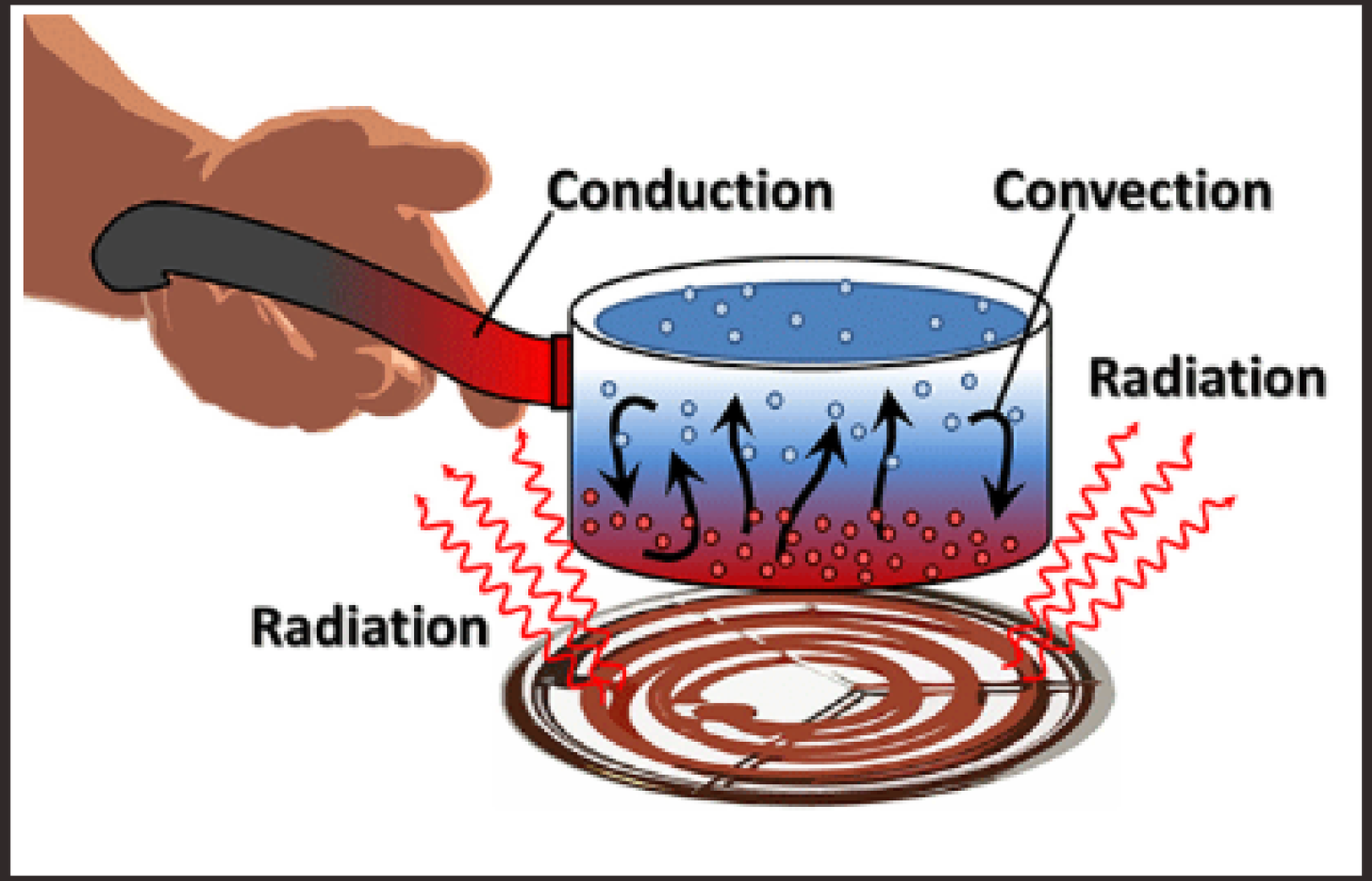
HOT Object (lost heat energy)

Q(lost)

COLD Object (gain heat energy)

Q(gained)

Heat Transfer : Three Methods



Heat Transfer Methods Explained

CONDUCTION
Direct Contact

✓ Metal spoon in soup
✓ Ironing clothes
✓ Hot pan handles

CONVECTION
Fluid movement

✓ Boiling water
✓ Sea Breezes
✓ Air currents

RADIATION
Electromagnetic waves

✓ Heat from Sun
✓ Solar panels
✓ Fire warmth

THANK YOU!